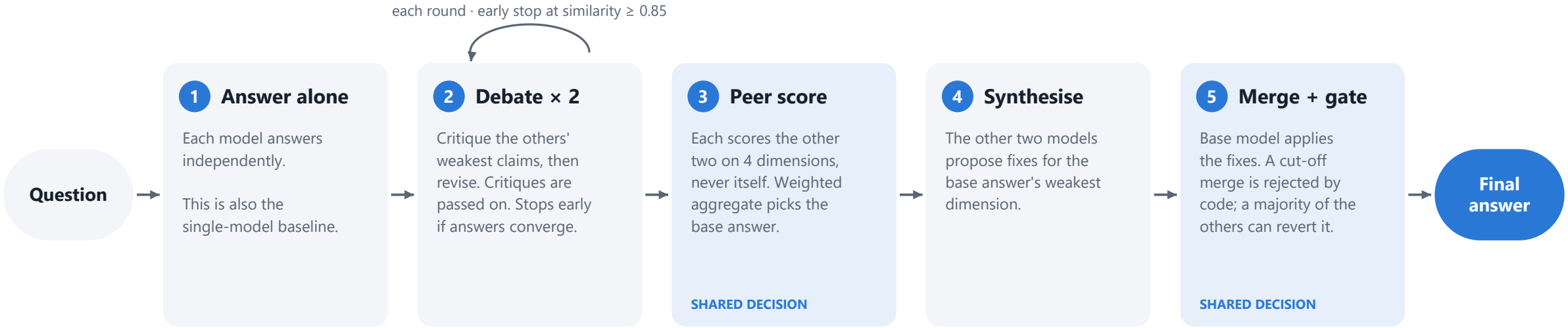
How it works: three models answer, argue, score each other and merge

No model ever picks or merges on its own — each decision is shared by several models or enforced by code



Debaters Llama 3.2 3B · Meta | Qwen 2.5 3B · Alibaba | Gemma 3 4B · Google — 3 models from 3 vendors, run via ollama

Evaluation — proving it was worth doing (runs separately; not part of producing the answer)

Baseline	Debate answer	Independent judges	Verdict
each model's step-1 answer	the final answer above	Phi-4 mini (Microsoft) + Ministral 3 3B (Mistral)	debate vs best single model

No judge shares a vendor with a debater. The judges are tested too: deliberately damaged answers must score lower (slide 2).

Guardrails: hard $25 spend ceiling in code · keys from environment variables only · one command to run · dry-run mode at zero cost · full transcript of every exchange

What “best” means — and a test that our judges can tell

Best = highest average rubric score from an independent judge panel; peers score with the same rubric

The rubric — each dimension 0–10

Completeness
Covers every part of the question
weight 25%

Accuracy
Claims are correct and supported
weight 25%

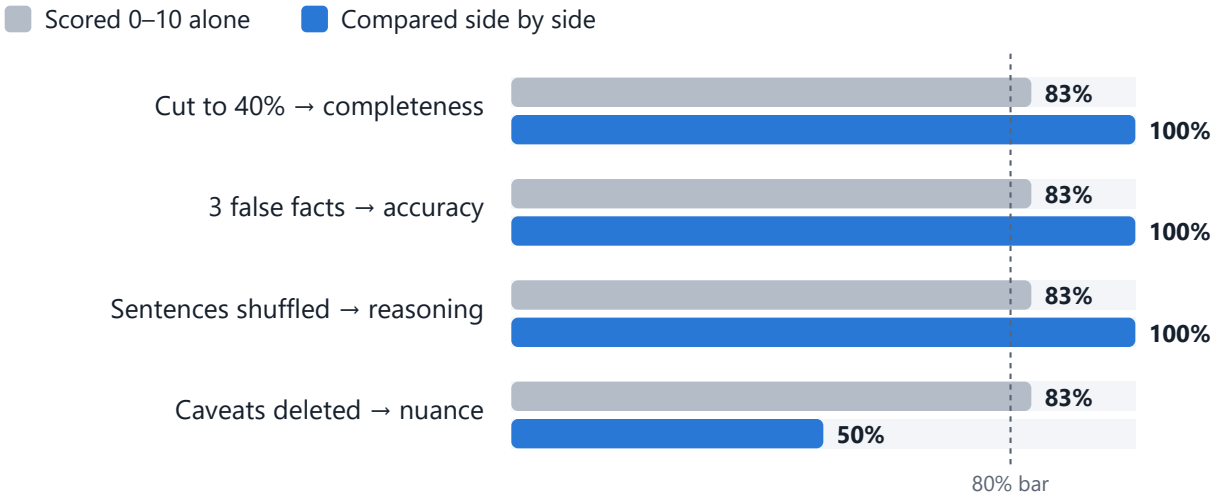
Reasoning
Logic is clear, structured and sound
weight 25%

Nuance
Handles trade-offs, edge cases, other views
weight 25%

Why not a vote? Three paragraphs are never “the same answer”, so voting cannot pick one. Scoring each dimension compares free text without forcing agreement.

The test: damage real answers, check the judges notice

6 real answers (one per question category), each damaged four ways by code. Share of damaged copies the judges ranked **below** the original:



PASS Scored 0–10 alone: caught 83% of 24 damaged copies

PASS Compared side by side: caught 88% of 24 damaged copies

Side-by-side judging passes, so it carries the evidence on slide 3.

Weak spot: caveats deleted caught 3/6. Losing caveats is the hardest damage for these judges to see.

Two judges beat one: on 0–10 scores the panel caught 83%; each judge alone caught 71% and 75%.

On 6 questions, the debate answer did not beat the best single model

Question set: 6 open-ended questions built for this test, one per category (pilot) · real run, local models

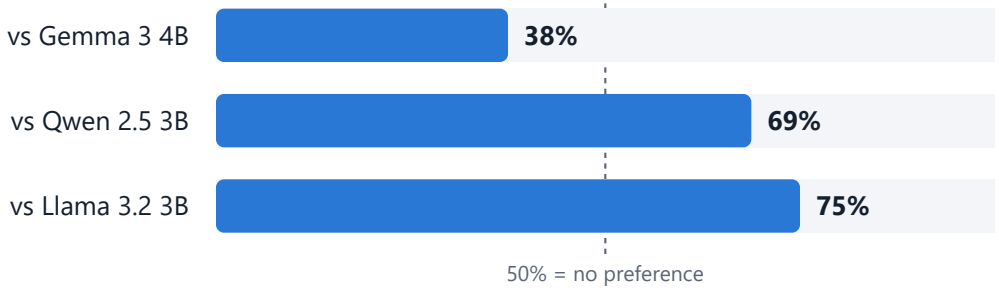
6
questions in the set

1 – 2 – 3
wins – ties – losses vs Gemma 3 4B, side by side

38%
of judgements preferred the debate answer (50% = even)

0.62
significance, sign test p (needs < 0.05)

How often judges preferred the debate answer, side by side



What this shows

- Judged side by side (the validated method), both orders, two independent judges.
- Against the toughest single model (Gemma 3 4B) the debate answer won 1, tied 2 and lost 3 of 6 questions.
- Against the weaker models it won 5 of 6 vs Llama 3.2 3B and 5 of 6 vs Qwen 2.5 3B.
- **Why not the best model:** the debaters chose Gemma 3 4B's answer to build on in only 1 of 6 questions, and won 1 of those. Small models are poor at spotting the strongest answer (fix: slide 6).
- On the 0–10 rubric: debate **8.90** vs best single model Gemma 3 4B **8.96**.
- 6 questions cannot show significance; the full 30-question run is one command.

Why a professional services firm would want this

Advisory work is open questions with no single right answer — exactly where one model's view is a risk

Built-in second opinion

Models challenge each other's weakest claims before anything reaches a reviewer, so one model's blind spot is caught by another.

An audit trail by default

Every answer ships with the full exchange: who claimed what, who challenged it, and how each answer scored. Fits review and QA culture.

Client data can stay in-house

This build runs entirely on local open-weight models: no data leaves the machine and no per-call fees. Cloud models are a config switch.

No single-vendor dependency

Three vendors debate. If one model degrades or is withdrawn, change one line of config; the method stays the same.

Where it fits: due-diligence memos · regulatory and policy analysis · risk assessments · first drafts of advisory positions — any question where a reviewer would otherwise ask “what did we miss?”

Measuring the return — and when to kill it

Baseline: the best single model answering alone, which is what teams do today

19	5.2 min	49k	~$0.20
model calls per answer (baseline: 1)	median per answer on a 6 GB laptop GPU	tokens per answer	per answer on cloud models (local: $0)

Measured in the evaluation run. Cloud figure applies Bedrock list prices (Claude Sonnet 4.5, Llama 3.3 70B, Mistral Large) to the measured tokens.

Measure on real client questions, against the baseline

Quality	reviewers pick debate vs baseline answer, blind
Reviewer time	minutes of edits before sign-off
Rework	revision cycles before the draft is accepted
Cost, latency	$ and minutes per answer, from the run log

Return = reviewer hours saved × rate − (model cost + waiting time).

Kill it if, after a 4–6 week pilot…

- reviewers prefer it in under 55% of blind comparisons
- reviewer edit time does not fall by at least 20%
- cost per answer exceeds the reviewer time it saves
- latency rules it out of the workflows that need it
- the judges cannot pass validation

With two more weeks — and where the model was wrong

Priorities come straight from what the testing showed

1. **Stronger debaters**
   3–4B models misjudge which answer to build on. Rerun on 70B-class or frontier models: about $0.20 per answer at Bedrock prices for the measured tokens; keep local models for confidential work
2. **Build on the best answer**
   stronger peer scoring, or merge the best parts of all three answers instead of editing one base
3. **Enforce the answer format**
   when a model skips the REVISED ANSWER marker, ask again instead of letting its critique through (this cost one question)
4. **Stronger, calibrated judges**
   frontier-model judges, calibrated on ~50 human-rated answers; close the blind spot for lost caveats (slide 2)
5. **Full and real question sets**
   all 30 built questions, then anonymised real client questions

Where the model was wrong, and I overrode it

The AI set the output limit to 1,024 tokens. The first real run then showed the debate losing, 7.38 vs 8.75. The transcript showed why: 4 of 6 debate turns and the merged answer were cut off mid-sentence. The loss was a bug, not a finding.

The models approved it. Both gate voters ACCEPTED a truncated 469-character answer.

Override: raised the limit to 2,048, made every call report truncation, and added a code guard so a cut-off merge can never replace the base answer; no model gets a vote on that. Rerun: no truncation, and the debate tied the best model on that question.